

Do Mushrooms Communicate Better Than Humans?

The surprising electrical signals travelling through fungal networks

Author: Kenisya Saravenen | Biomedical Sciences Undergraduate

What if the next time you walked past a mushroom, it was communicating with the fungi around it?

It sounds like something from science fiction, but fungi really do produce electrical signals. Unlike humans, fungi do not have brains or nervous systems. Instead, many fungi grow as networks of microscopic, thread-like structures called hyphae, which collectively form a larger network known as a mycelium. These interconnected networks allow fungi to grow through their environment and transport materials throughout the colony.

Researchers have discovered that these fungal networks also produce patterns of electrical activity. A 2022 study by Andrew Adamatzky recorded electrical signals from four fungal species: ghost fungus, enoki fungus, split-gill fungus and caterpillar fungus. Tiny electrodes were used to measure changes in electrical potential within fungal material.

So... are fungi actually talking?

The recordings revealed repeated bursts, or spikes, of electrical activity. These spikes were not simply random fluctuations: they could occur in groups or trains, with different patterns observed between species. The signals also appeared to propagate through fungal networks, suggesting that electrical activity could potentially transmit information between different parts of a colony.

This is where the research becomes particularly intriguing. Adamatzky analysed the electrical spike patterns using methods inspired by linguistics. Groups of spikes were treated as “words”, while larger sequences were considered “sentences”. Based on this approach, the study estimated a possible fungal lexicon of up to around 50 signal patterns, although the most frequently occurring patterns were much fewer, generally around 15–20.

The lengths of these proposed fungal “words” showed statistical similarities to word lengths found in human languages. *Schizophyllum commune*, commonly known as split-gill fungus, produced the most complex sequences among the four species studied.

A crucial scientific caveat

The study did not demonstrate that individual fungal signals have specific meanings such as “danger” or “food”. The term “language” is an interpretation based on statistical analysis of electrical patterns. The biological function of these signals remains uncertain and is still an active area of research.

What could these signals actually do?

One possibility is that electrical activity forms part of the way fungi coordinate activity across their networks. Fungal colonies continually interact with their environment, respond to changes and distribute resources through their interconnected hyphae. Electrical signalling could potentially contribute to this coordination.

However, there is still a lot we do not know. The most interesting aspect of this research may therefore be less about discovering a literal “mushroom language” and more about challenging our assumptions about where biological information processing can occur.

We often associate electrical signalling with neurons and the brain, yet electrical activity also occurs in organisms without nervous systems. Fungi provide an unusual example of how living systems can generate, transmit and organise electrical signals without a conventional nervous system.

Perhaps fungi are not “talking” in the human sense. But their electrical activity suggests that communication and information processing in living organisms may be much more diverse than we once thought.

Maybe mushrooms are not better communicators than humans, maybe we have simply been listening in the wrong way.

The Science at a Glance

Fungi studied	4 species: ghost, enoki, split-gill and caterpillar fungi
Signal measured	Extracellular electrical potential
Recording method	Differential electrodes inserted into fungal material / colonised substrate
Key finding	Electrical spikes often occurred in organised trains with species-specific patterns
Proposed interpretation	Potential information transmission through fungal networks
Important limitation	No specific meaning has been established for individual signals

Further Reading

Adamatzky, A. (2022). Language of fungi derived from their electrical spiking activity. Royal Society Open Science, 9(4), 211926. DOI: 10.1098/rsos.211926.